21. Friday, May 16 **Indexed Families of Sets**

Theorem 1: Let Λ be a non-empty index set. Let $\{A_\alpha \mid \alpha \in \Lambda\}$ be an indexed family of sets. Then:

1) For each $\beta \in \Lambda$, $A_\beta \subseteq \bigcup_{\alpha \in \Lambda} A_\alpha$

2) For each $\beta \in \Lambda$, $\bigcap_{\alpha \in A} A_\alpha \subseteq A_\beta$

3) DeMorgan's laws: $\left(\bigcup_{\alpha \in \Lambda} A_\alpha\right)^c = \bigcap_{\alpha \in \Lambda} A_\alpha^c$ and $\left(\bigcap_{\alpha \in A} A_\alpha\right)^c = \bigcup_{\alpha \in \Lambda} A_\alpha^c$

Proof: Let's proof the second DeMorgan's law. We prove both inclusions.

[$\subseteq$] Let $x \in \left(\bigcap_{\alpha \in \Lambda} A_\alpha\right)^c$. Then $x \notin \bigcap_{\alpha \in \Lambda} A_\alpha$. Since x is <u>not</u> in the intersection, by def^n this means $\exists\, \beta \in \Lambda$ such that $x \notin A_\beta$. Thus $x \in A_\beta^c$, Therefore $x \in \bigcup_{\alpha \in \Lambda} A_\alpha^c$ by definition.

[$\supseteq$] Suppose $x \in \bigcup_{\alpha \in \Lambda} A_\alpha^c$. By def^n, $\exists\, \beta \in \Lambda$ such that $x \in A_\beta^c$. Hence $x \notin A_\beta$.

By definition of intersection, $x \notin \bigcap_{\alpha \in \Lambda} A_\alpha$. Hence $x \in \left(\bigcap_{\alpha \in \Lambda} A_\alpha\right)^c$

The other statements are exercises $\square$

Functions (IntVo, 6.1 and 6.2)

Definition 2 (Version 1) Let A and B be sets

A function $f: A \longrightarrow B$ is a rule that assigns to each $a \in A$ exactly one element $f(a) \in B$.

(:colon, \to, \rightarrow, \longrightarrow)

We say A is the **domain** and B is the **codomain** of f.

Notation: $f: A \longrightarrow B$ — We read "f is a function from A to B".

f: name; A: domain; B: codomain

Definition 2 Let $f: A \longrightarrow B$ be a function.

- If $a \in A$, the element $f(a) \in B$ is called the **image of a** under f.
- If $X \subseteq A$, the subset $f(X) = \{f(a) \mid a \in X\} \subseteq B$ is called the **image of X** under f.
- If we take all of A, $f(A) = \{f(a) \mid a \in A\} \subseteq B$ is called the **range of f** or the **image of f**.

Can be smaller than B, the codomain.

Example: ~~$f(x) = x^2$~~ Let $f: \mathbb{R} \longrightarrow \mathbb{R}$ defined by $f(x) = x^2$.

Here domain $= \mathbb{R}$ and codomain $= \mathbb{R}$.

Range of $f = [0, \infty) \neq$ codomain of f

And $f((-2, 2]) = [0, 4]$

Q: Why don't we define f with codomain $[0, \infty)$?

It makes sense to keep codomain larger when operating with several functions.

We can also define $g: \mathbb{R} \longrightarrow [0, \infty)$ by $g(x) = x^2$.

But $f \neq g$, because the codomains are different.

Even more contrasting example:

Take $h: \mathbb{Z} \longrightarrow \mathbb{Z}$ given by $h(x) = x^2$. Then $h \neq f$ and $h \neq g$

Because domains and codomains are different.

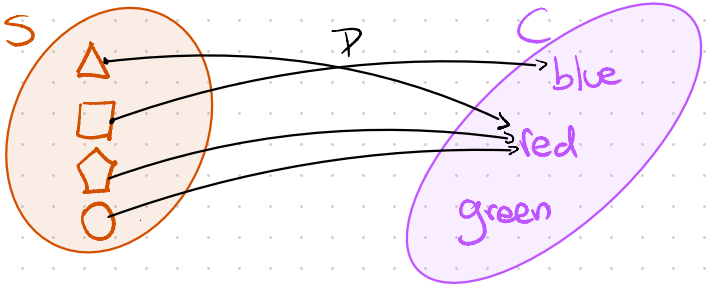
Example: (Can define functions without using algebraic formulas.)

Consider the sets $S = \{\triangle, \square, \pentagon, \bigcirc\}$ and $C = \{$blue, red, green$\}$

We define a function $p: S \to C$ by

$p(\triangle) =$ red, $p(\square) =$ blue, $p(\pentagon) =$ red and $p(\bigcirc) =$ red

This is a well-defined function. Here is a picture:



Here

Range of $p = \{$blue, red$\}$

Definition 3: Let $f: A \to B$ and suppose $b \in B$ and $Y \subseteq B$.

- A **preimage of** b (under f) is an element $a \in A$ such that $f(a) = b$.

  Note: b may have non or many preimages.

- The **preimage of** Y is the set of all preimages of all elements of Y. That is,

$$f^{-1}(Y) = \{a \in A \mid f(a) \in Y\}$$

  ↳ We are not saying f is invertible. Can use $f^{-1}[Y]$ to emphasize $Y \subseteq B$.

Example (continued) In the previous example:

- green has no preimage
- □ is a preimage of blue
- $p^{-1}(\{\text{red}\}) = \{\triangle, \square, \bigcirc\}$

Injections, Surjections, Bijections (6.3)

Definition 4: A function $f: A \to B$ is called **injective** if

> For all $x, y \in A$, if $f(x) = f(y)$ Then $x = y$

"An injection maps different Things To different Things"

Equivalent (contrapositive)

> For all $x, y \in A$, if $x \neq y$ Then $f(x) \neq f(y)$

In other words, $f: A \to B$ is injective if for each $b \in B$, There exists at most one $a \in A$ such That $f(a) = b$.

Technique: Let $f: A \to B$

- To prove f is injective, assume some $x, y \in A$ satisfy $f(x) = f(y)$, and prove That $x = y$.
- To prove f is not injective, find some $x, y \in A$ with $x \neq y$ but $f(x) = f(y)$.

Example: Back to previous shapes and colors example. The given function is not injective

Because $\triangle \neq \pentagon$ but $p(\triangle) = \text{red} = p(\pentagon)$

In these example, an injection $q: S \longrightarrow C$ would give a way to color each shape with a different color. That's not possible.

Examples:

1) $f: \mathbb{R} \longrightarrow \mathbb{R}$, $f(x) = x^2$ is not injective: because $f(-1) = f(1)$ but $-1 \neq 1$.

2) $g: \mathbb{R} \longrightarrow \mathbb{R}$, $g(x) = \cos x$ is not injective: $\cos(0) = \cos(2\pi) = \cos(4\pi) \ldots$ but $0 \neq 2\pi$

3) $h: \mathbb{R} \longrightarrow \mathbb{R}$, $h(x) = 2$ is not injective, $h(0) = h(1)$ but $0 \neq 1$.

4) $\ell: \mathbb{R} \longrightarrow \mathbb{R}$, $\ell(x) = 2x$ is injective.

In fact, if $x, y \in \mathbb{R}$ are such that $\ell(x) = \ell(y)$, then $2x = 2y$, and multiplying by 2^{-1} we get $x = y$.

Definitions: A function $f: A \to B$ is called **surjective** if for each $b \in B$ there exists some $a \in A$ so that $f(a) = b$.

In other words, range of f = codomain of f "A surjection hits all elements of the codomain"

Technique: Let $f: A \to B$

- To prove f is surjective, fix an arbitrary $b \in B$ and construct some $a \in A$ such that $f(a) = b$.
- To prove f is not surjective, find some $b \in B$ such that $f(a) \neq b$ for all $a \in A$.

Example: The function $p: S \to C$ from before is not surjective, because green is not in the range. In other words, for each $s \in S$, we have $p(s) \neq$ green.

Examples:

1) $f: \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$ is not surjective because for any $x \in \mathbb{R}$, $f(x) \geq 0$, so $f(x) \neq -1$.

2) $\ell: \mathbb{R} \to \mathbb{R}$, $\ell(x) = 2x$ is surjective.

In fact, given any $b \in \mathbb{R}$, if we take $a = 2^{-1}b$, then $\ell(a) = 2a = 2 \cdot 2^{-1}b = b$.